

Hospital & Patient Analytics

Data Story & Analysis Report

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Executive Summary

Across 10,000 unique patient records (after removing 15 duplicate entries from the raw data), this hospital dataset shows KES 172,668,300 in total billing between 2023 and September 2026, an average bill of KES 17,267 per patient, and an average stay of 3.77 days. This report walks through what drives cost, where the data is reliable, and one gap worth closing before the next reporting cycle.

10,000 Total Patients	KES 172.7M Total Billing	KES 17,267 Avg. Billing/Patient	3.77 days Avg. Length of Stay
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1. What Drives Cost: Length of Stay, Not Diagnosis Severity

Average billing climbs steadily and consistently alongside average length of stay across every diagnosis — this is the central finding of the analysis. Depression and Migraine sit at the bottom of the cost list (KES 15,921 and KES 15,925, averaging about 3.3 days), while Pneumonia, COPD, and Appendicitis sit at the top (KES 17,927–KES 18,836, averaging 3.9–4.3 days). The pattern is not random: costlier diagnoses are consistently the ones that keep patients admitted longer, which is the expected and reassuring explanation rather than a data anomaly.

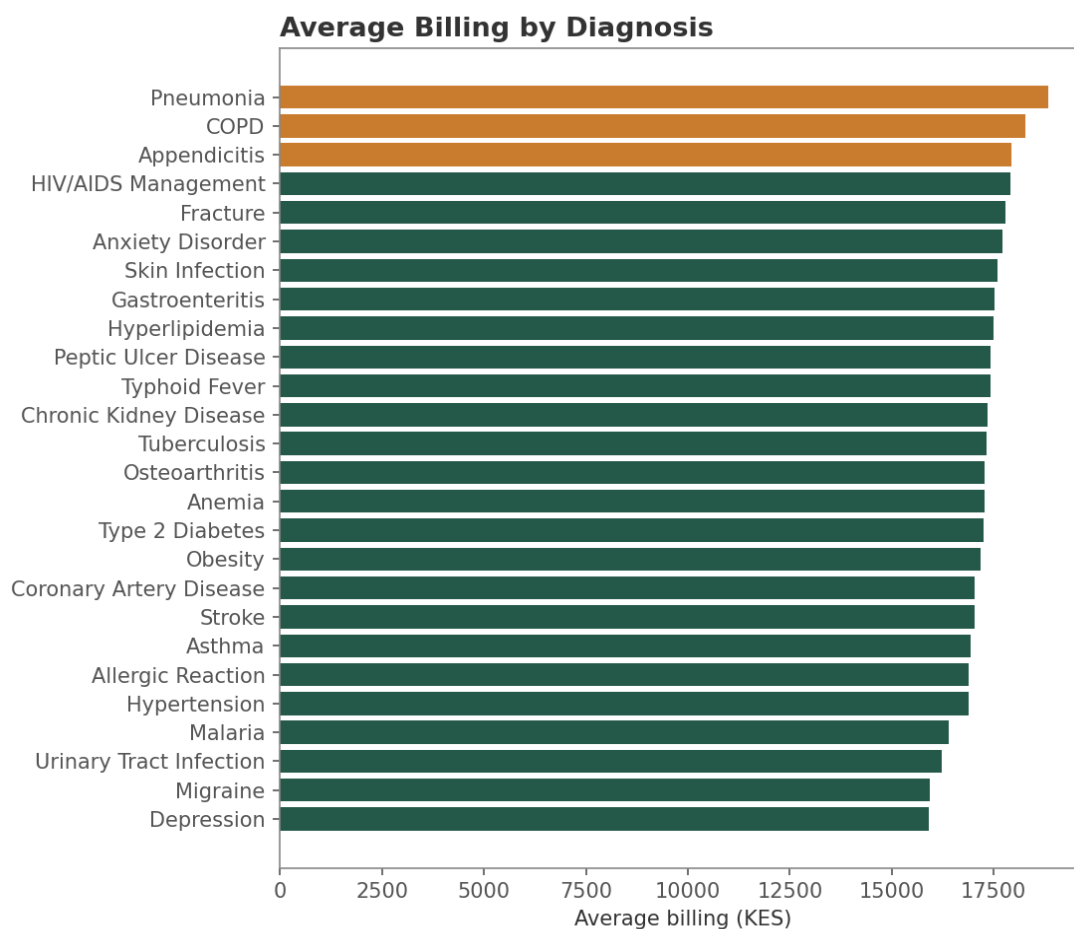


Figure 1. Average billing by diagnosis, sorted low to high. The three highest-cost diagnoses (highlighted) also have the longest average stays.

Patient volume is very even across diagnoses (367–411 patients each), so no single condition is over-represented enough to skew these averages — the cost differences reflect genuine clinical patterns, not a sampling artifact.

2. Admission Type: Similar Stays Across the Board

Average length of stay is fairly consistent across all four admission types, ranging narrowly from 3.65 days (Routine Checkup) to 3.87 days (Urgent). Elective admissions are the most common (3,519 patients), followed by Emergency (3,005), Urgent (2,021), and Routine Checkup (1,455).

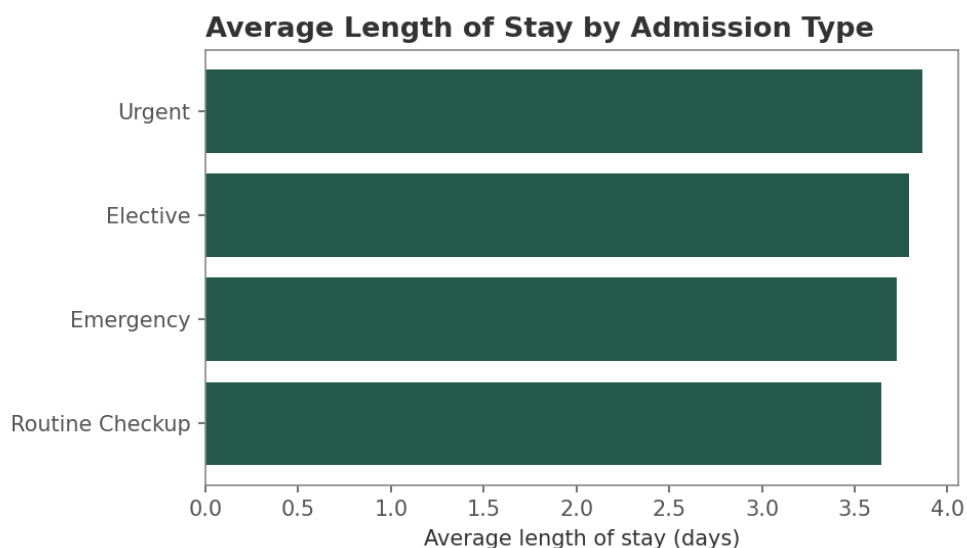


Figure 2. Average length of stay by admission type — no type stands out as dramatically more resource-intensive.

This is a useful negative finding: hospital resourcing does not need to treat any one admission type as a structurally longer or costlier category than the others.

3. Yearly Trend: Flat, With a Partial Final Year

Admissions run close to 2,700 per year across 2023–2025 (2,660 / 2,705 / 2,730), with no meaningful growth or decline. The 2026 figure of 1,905 admissions looks lower only because the year's data runs through September, not because volume is falling — at the same monthly pace, 2026 is on track to land close to prior years. Average billing per patient is similarly flat, staying within a tight KES 16,981–KES 17,612 band across all four years, meaning there is no cost inflation trend to report.

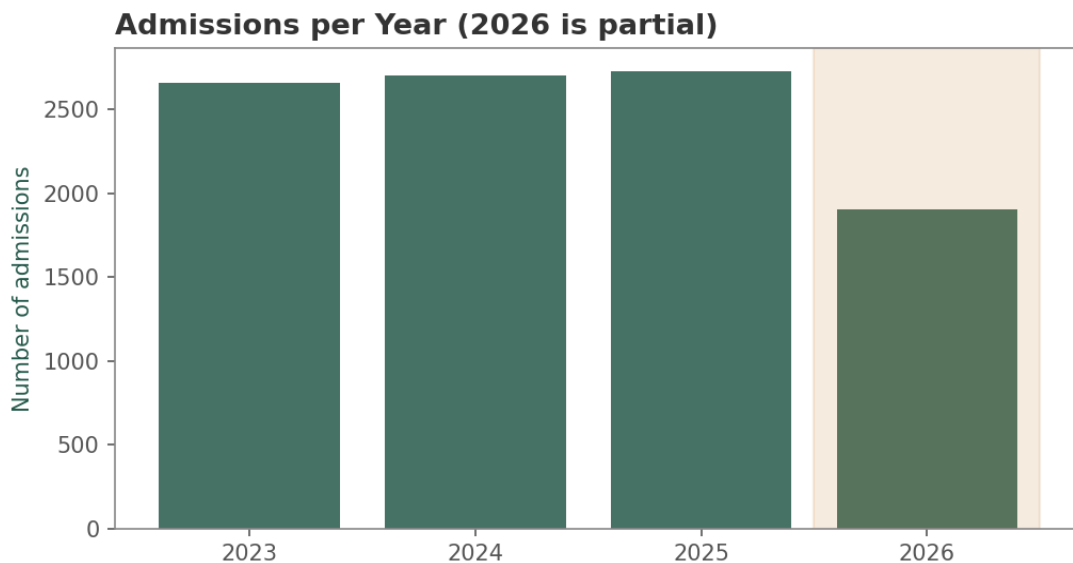


Figure 3. Admissions per year. The 2026 bar (highlighted) reflects a partial year, not a real decline.

4. Insurance Coverage

NHIF/SHA is by far the most common payer, covering 2,943 patients — more than double the next largest provider (AAR Insurance, 1,204). Self-Pay accounts for 830 patients, and 200 records have no insurance provider recorded at all.

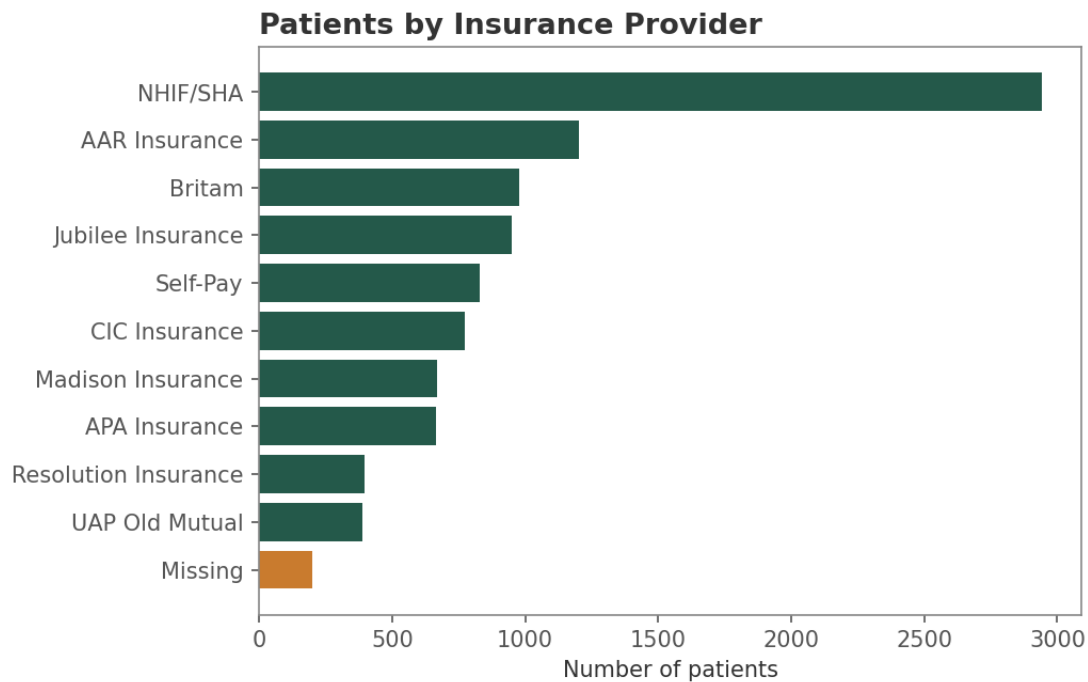


Figure 4. Patients by insurance provider. The highlighted bar marks records with no provider recorded.

5. Test Results

Just over half of all records (53.9%) show a Normal test result, 33.9% show Abnormal, 9.2% are Inconclusive, and 3.0% (300 records) have no test result recorded at all.

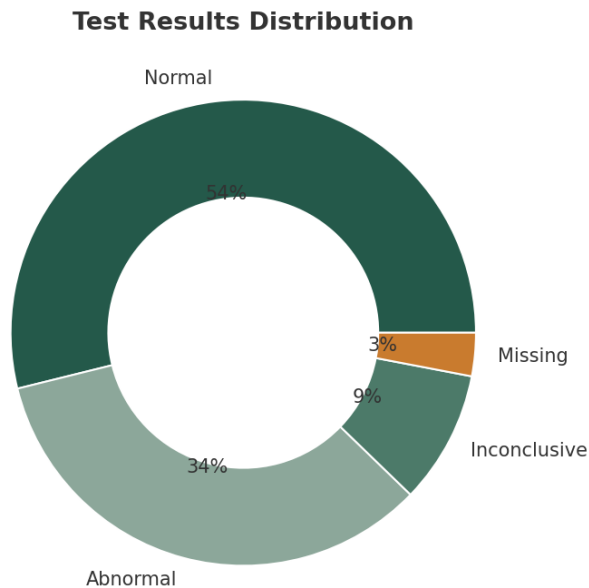


Figure 5. Test results distribution across all patients.

6. Data Quality Note

The raw dataset contained 15 exact duplicate records, which were removed before any of the figures above were calculated. Two fields still carry missing values after cleaning: Test_Results (300 records, 3.0%) and Insurance_Provider (200 records, 2.0%). These gaps were preserved rather than filled with assumed values, so all percentages and averages in this report reflect only the records where that specific field was recorded — the true completion rate would need those 500 records followed up on to close.

Recommendations

- Use length of stay, not diagnosis type alone, as the primary driver when forecasting or budgeting for patient costs — the relationship is consistent and predictable.
- When reporting 2026 figures, always note the year is partial (through September) to avoid the lower admission count being misread as a genuine decline.
- Follow up on the 200 missing Insurance_Provider records, since correctly attributing payer mix matters for billing and reimbursement reporting.
- Investigate the 300 missing Test_Results records to determine whether they reflect pending tests, untested patients, or a data-entry gap — each implies a different fix.
- Continue monitoring admission-type mix; while stay lengths are currently balanced, Elective's high share (35% of admissions) is worth tracking if scheduling capacity becomes constrained.

Methodology: Data was de-duplicated (15 exact duplicate rows removed) before any aggregation. All averages and totals reflect the 10,000 remaining unique patient records. Missing values in Test_Results and Insurance_Provider were left as missing rather than imputed, and are reported as a distinct category where relevant.